

Light-Weight Korean Lip-Sync - Project Proposal

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Index Terms—lip-sync, phoneme, viseme, GAN, mel-spectrogram.

I. INTRODUCTION

A. Problem Statement

Many generative lip-sync AI models, including the widely used Wav2Lip [1], have been trained and evaluated on English data. More recent work shows that performance degrades when models are extended to languages that are phonetically distant from English [2]. However, the field has not established whether closing this gap requires new cross-lingual architectures or if computationally light retraining is sufficient [4], [5]. This project will evaluate this directly, using Korean - a language with historically limited audio-visual lip-sync resources [8] and phonetic structure very different from English - as a test case.

B. Motivation

Language is the main barrier to cross-culture media access. Dubbing is an imperfect substitute when lip movements visually don't match spoken audio. Accurate cross-lingual lip-sync would improve the accessibility of foreign language films, while also having adjacent applications in medical communication, hearing accessibility, and translated court testimony.

C. Goals and Objectives

This project seeks to evaluate whether computationally light, language specific, fine-tuning of an existing English lip-sync model is sufficient to close the cross-lingual performance gap, using Korean as the target language. The three key steps I will follow are: (1) baseline evaluation, (2) generator fine tuning, (3) comparative evaluation against reported results for recent existing cross-lingual approaches like MuEx [2] and [3]. If time permits, a phoneme-level diagnostic metric will be developed to further investigate the failures [6].

II. BACKGROUND AND LITERATURE REVIEW

Wav2Lip [1] is my baseline model that I will evaluate and retrain in a Korean context. It uses a unique approach, where the discriminator is an already pretrained “lip-sync expert” which can distinguish in-sync from out-of-sync audio-video pairs. This expert is based off the work of SyncNet [11], using Lip-Sync Error - Distance (LSE-D) and Lip-Sync Error - Confidence (LSE-C) as its evaluation metric. This evaluation allows the Wav2Lip model to stably iterate toward a more accurate lip-sync parameters during training, as opposed to training the discriminator alongside the generative model.

Wav2Lip's generator minimizes:

$$L_{\text{total}} = (1 - s_w - s_g) \cdot L_{\text{recon}} + s_w \cdot E_{\text{sync}} + s_g \cdot L_{\text{gen}}$$

combining L1 pixel reconstruction, a frozen sync-expert loss, and an adversarial visual-quality loss [1]. The sync-expert discriminator uses a fixed $T_v = 5$ video frame window; Wav2Lip's own study found that larger window sizes consistently improved lip-sync discrimination accuracy, but capped it at 5 frames. CALS [10] extended this same trend, showing lip-sync quality continues to improve up to an approximately 1.2 second window, which suggests Wav2Lip's short-window design leaves phonetic context unused.

Both Oskooei et al. [4] and Venkataraghavan et al. [5] evaluated Wav2Lip cross-lingually - the former on Turkish, Persian, and Arabic, the latter adapting it for Indian languages. However, both approaches rely solely on Wav2Lip's built-in SyncNet LSE-D/LSE-C metrics. This is a limitation, since these metrics are trained on English audio-visual data, they may carry an English-centric bias rather than measuring true phoneme/viseme correspondence in the target language.

Heavier architectures like MuEx [2] and DiffDub [3] have also emerged to address cross-lingual lip-sync. MuEx develops a phoneme-guided mixture-of-experts for phoneme/viseme mapping, at greater architectural cost than Wav2Lip's single generator. Unlike Wav2Lip's GAN-based generator, DiffDub uses a diffusion model that iteratively de-noises the lower-face region based on audio. This avoids GANs training instability while leaving the rest of the frame (background, pose) untouched [3]. This is an entirely different paradigm (iterative de-noising vs adversarial training) that makes the process layers more complex than Wav2Lip's light-weight generation.

Phoneme/viseme-level modeling has been found to strengthen generative lip-sync quality. Both SyncTalkFace and CALS demonstrate the idea of phoneme-aware and context-aware representations that meaningfully improve the sync quality over Wav2Lip's short audio-window approach [9], [10]. This is the exact reason why I want to explore a phoneme-level diagnostic metric to truly evaluate the quality of my fine-tuned Wav2Lip model. Focusing more on the linguistic context is a better evaluation heuristic than just a distance/confidence sync, as language context is what will make the sync look more “realistic” to viewers.

Tapu et al. [7] point out that there is a resource imbalance with English/non-English data in the related task of visual speech recognition, and that is the reason why most models are more effective in English (and other widely used languages).

This same imbalance extends to lip-sync generation: less common languages, including Korean, are under-represented with data, which is why cross-lingual models aren't as effective as English-based models. Fortunately for my project, I found OLKAVS [8], the largest open-source collection of Korean visual speech data intended to fill this data hole. I will be using a subset of this dataset to finetune Wav2Lip.

So far, there has been no direct research into light-weight retraining closing the cross-lingual gap specifically in Korean, nor whether phoneme-level diagnostics could be repurposed into an evaluation tool instead of a training signal.

III. DESIGN

Baseline and Proposed Method. The baseline for this project is the current unadapted Wav2Lip model. I will preemptively test Wav2Lip on the same test set I will use for my fine-tuned version of the model. As described in Section II, L_{total} is a weighted-total loss between L_{recon} , E_{sync} , and L_{gen} . For the purposes of fine-tuning the model, I will only retrain the generative part of the model, freezing the SyncNet-based sync-expert discriminator. This mirrors Wav2Lip's own findings that fine-tuning the discriminator on generated frames degrades its ability to detect true audio-lip correspondence. Freezing it also means that I will be using the same, unchanged metric scores in both the baseline and fine-tuned model, giving me a consistent comparison. I acknowledge that, as stated in Section II, this metric remains English-biased, which is precisely why my stretch goal is to develop a phoneme-level diagnostic metric to act as a complimentary, language agnostic evaluation tool.

Dataset and Data Processing. For my Korean dataset I will be using a subset from OLKAVS [8]. The subset will be sized at roughly five-hundred clips, and each clip only being between two and ten seconds each. The size was chosen for feasibility - a subset this small should train in a few hours on an RTX3080.

The Wav2Lip [1] input requires me to design a pre-processing pipeline to run all of my OLKAVS data. Wav2Lip defaults to 25fps, which I will need to update to 30fps to match OLKAVS's rather than resampling, which is a non-integer (6:5) ratio, risking the introduction of timing artifacts and audio-visual misalignment. Wav2Lip's pipeline uses S3FD for face/lip detection; however, OLKAVS already provides face and lip bounding boxes, which I will use directly if sufficient, avoiding the need to re-run detection. This will need to be resized to 96x96 to match Wav2Lip's requirement. The audio will also need to be downsized to 16kHz mono and converting to an 80-channel mel-spectrogram ($n_{fft}=800$, $hop_size=200$, $win_size=800$) to match the exact parameters the frozen SyncNet expert was trained on.

The English dataset I will use will be LRS2, since that is what Wav2Lip was originally trained/evaluated on, to keep things simple [1].

Software, Hardware, Codebase. My codebase will be built from Wav2Lip's public GitHub repository, using Pytorch. For the computation, I will be using my own personal RTX3080

to run all the training and store data, and use Kaggle's free T4 GPU as a backup if that fails.

Experiments and Evaluation Measures. The first step of the experiment is to evaluate the base Wav2Lip model on my Korean test set. This will first establish LSE-D/LSE-C degradation gap. Next, I will fine-tune the generator on my Korean training set, and re-evaluate on both Korean and English test sets using LSE-D/LSE-C. It is important to evaluate on an English test set to see if there is any model degradation after fine-tuning in Korean. After this, I will benchmark my Korean results against the architecture heavy MuEx [2] and DiffDub [3] models and evaluate whether a light retraining is more sufficient in a Korean context. If time permits, I will implement a phoneme-level diagnostic metric, using Montreal Forced Aligner to detect phonemes. I will bucket these phonemes with my LSE-D/LSE-C errors and can evaluate where exactly my model goes wrong in the Korean language.

Ethical and Feasibility Considerations. With research in this field there is a large concern with deepfakes and spreading misinformation. Wav2Lip's own paper caution that any output generated using their model should unambiguously present itself as synthetic [1]. This project follows this guidance directly, using OLKAVS consenting, pre-cleared speaker data [8] rather than public figures, with all generated outputs clearly labelled as synthetic.

Some things can go wrong with my timeline, data pre-processing could take longer than expected, and training could be delayed. This is mitigated by using such a small subset, as I am able to train my model in a few hours, any errors I run into won't put me further behind than a few days.

IV. MILESTONES AND PROJECT TIMELINE

TABLE I
PROJECT TIMELINE

Date	Milestone
11/08	Preprocess OLKAVS subset (crop, resample, mel-spectrogram)
1/09	Evaluate unadapted Wav2Lip baseline on Korean and English test sets
10/09	Fine-tune Wav2Lip generator on Korean subset
24/09	Re-evaluate fine-tuned model on Korean and English test sets (LSE-D/LSE-C); compare against baseline and reported results [2]–[5]
29/09	<i>If time permits:</i> develop and apply phoneme-level diagnostic metric
06/10	Final report writing, presentation preparation

The table above shows my deadlines for each part of the project. They are mainly in line with the mandatory check-in sessions that are outlined in the lecture schedule. I have overestimated the time it would take for each milestone, to account for errors, as well as giving myself over a weeks buffer with my stretch goal and final report/presentation deadline.

STATEMENT

I, James Brooker, state that the research, writing, and idea's behind this report are unambiguously my own. This report was written using LaTeX. Generative AI was used to assist with

researching parts of the methodology (what needs to be done in a pre-processing pipeline), and as a writing structure/grammar checker. The writing was originally written by me, and I used generative AI to grade and give feedback on it, without rewriting it. The voice, knowledge, and idea's shown are my own.

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